

Michael Georgariou III

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EXPERIENCE

• Apple

Cupertino, CA

Software Engineer - Embedded Debug Tools

May 2025 - Present

- Overhauled a system daemon brokering hardware debug probe access, moving it to a smart-pointer ownership model and shifting device discovery and controller state onto dispatch queues, fixing over a dozen field-reported crashes from races and use-after-free bugs
- Built a new C library from scratch for host-to-probe communication: USB transport, device protocol, firmware recovery and reset, authentication, and chip identity APIs
- Added cross-triggering support for on-chip debug hardware, including a reusable base class, alarm handling, and a clean public C++ and Tcl API
- Stood up my team's first CI system on Jenkins: builds, artifact publishing, automated pull request status comments, log capture, and change-gated tests
- Delivered debug and bring-up support for several pre-release systems-on-chip, covering configuration engines, address translation hardware, and processor memory maps
- Extended LLDB and GDB debugging across four processor architectures with target register descriptions and physical memory access, and improved core dump capture for early-boot panics
- Modernized the logging subsystem, retiring legacy interfaces and adding user-directed log files

• Amazon

San Diego, CA

Software Development Engineer II

Jan. 2022 - May 2025

- Re-ranked widgets across Amazon with a new returns-data algorithm, generating millions of daily impressions and reducing items sold and later returned
- Implemented a new badging framework which displayed a "Frequently Returned Item" badge on products in Amazon's catalog that exceeded return rate thresholds
- Owned and wrote several high- and low-level design documents, which interns and junior team members implemented and promoted to production from my specifications
- Led efforts to deprecate features with low customer utilization, letting the team pick up an additional story each sprint and cutting on-call burden

• Hewlett-Packard Enterprise (Aruba Networks)

Roseville, CA

Systems/Software Engineer

Jun. 2021 - Jan. 2022

- Created and led a prototype hardware support team to automate and simplify the process of working with in-office hardware remotely, supporting around 100 engineers
- Reworked a test framework and led a code refactoring effort for a new daemon being produced by my team

PROJECTS

• MPGameBoy

Written in C

Oct. 2022 - Present

- Implemented the Nintendo GameBoy's architecture in portable C code to load and read ROMs for the console
- Wrote CPU implementation, instruction handling, a debugger, and a tile-set viewer

• Minls and Minget

Written in C

Jun. 2021

- Created a filesystem reader for Minix, for use outside of the Minix operating system, in portable C code
- Supported functions to list out the contents of a directory and print the contents of a file

EDUCATION

• California Polytechnic State University

Bachelor of Science in Computer Engineering; GPA: 3.53

San Luis Obispo, CA

Aug. 2017 - Jun. 2021

PROGRAMMING SKILLS

Languages: C/C++, Python, Tcl

Architectures: ARM64, RISC-V, Cortex-M, Xtensa

Tools: Git, Unix, Xcode, LLDB/GDB, Jenkins

Concepts: JTAG/SWD, multithreading, USB